NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

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Master in Computer Science and Engineering

COST AND OPERATIONS OPTIMIZATION IN MULTI-CLOUD ENVIRONMENTS OVER GENAI FRAMEWORKS

MASTER IN COMPUTER SCIENCE AND ENGINEERING

SPECIALIZATION IN SPECIALIZATION NAME

NOVA University Lisbon

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Abstract

Keywords: Multi-Cloud · Generative AI · Cost Optimisation · Cloud Operations · GenAI Frameworks

Resumo

Palavras-chave: Multi-Cloud · Inteligência Artificial Generativa · Otimização de Custos · Operações em Nuvem · Frameworks GenAI

Disclosure of Delegation to Generative Artificial Intelligence

Duarte José Lourenço Belo declares[1] the use of Generative AI in the research work and writing process of this document. According to the GAIDeT taxonomy [15], the following tasks were delegated to Generative AI tools under full human supervision:

Conceptualization

- [x] Idea generation
- [x] Defining the research objective
- [x] Formulating research questions and hypotheses
- [ ] Feasibility assessment and risk evaluation
- [ ] Preliminary hypothesis testing

Literature Review

- [x] Literature search and systematization
- [x] Writing the literature review
- [ ] Analysis of market trends and/or patent environment
- [ ] Evaluation of novelty and identification of gaps

Methodology

- [x] Research design
- [x] Development of experimental or research protocols
- [ ] Selection of research methods

Software Development and Automation

- [x] Code generation
- [x] Code optimization
- [ ] Process automation
- [ ] Creation of algorithms for data analysis

Data Management

- [x] Data collection
- [x] Validation
- [x] Data cleaning
- [ ] Data curation and organization
- [ ] Data analysis
- [ ] Visualization
- [ ] Reproducibility testing

[1] This declaration was generated using LuaLaTeX [16] and the LaTeX package `aidisclose` [10].

Writing and Editing

- [x] Text generation
- [x] Proofreading and editing
- [x] Summarizing text
- [ ] Formulation of conclusions
- [ ] Adapting and adjusting emotional tone
- [ ] Translation
- [ ] Reformatting
- [ ] Press releases and outreach materials

Ethics Review

- [x] Bias analysis and discrimination assessment
- [x] Ethical risk analysis
- [x] Monitoring compliance with ethical standards
- [ ] Data confidentiality monitoring

Supervision

- [ ] Quality assessment
- [ ] Trend identification
- [ ] Identification of limitations
- [ ] Recommendations
- [ ] Publication support

Generative AI tools used: ChatGPT-5.2, Gemini 3 Pro, Claude 4.5 Opus, and Perplexity 12.12.25.

Additional comment #1: The AI was used primarily for refining the code in Section 3, but the algorithm logic was derived manually.

Additional comment #2: This is a second additional comment!

Additional comment: This is a third additional comment! It is starred and, thus, does not get a number. Also, this comment is longer and, hopefully, will span to the next line!

Additional comment #3: This is the third non-starred additional comment!

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Background

2.1 Foundations of Neural Language Representation

The ability to represent language in a form that machines can process and reason over is a prerequisite for any modern Generative AI system. Two landmark contributions established the theoretical and practical foundations upon which contemporary large language models (LLMs) and retrieval-augmented architectures are built: the introduction of dense word representations via neural networks, and the development of the Transformer architecture based solely on attention mechanisms.

2.1.1 Distributed Word Representations

Prior to 2013, natural language processing relied predominantly on sparse, high-dimensional representations such as one-hot encodings, which offered no intrinsic notion of semantic similarity between words. Mikolov et al. [11] addressed this limitation by proposing Word2Vec, a computationally efficient framework for learning continuous vector representations of words from large unlabelled corpora. The model is trained through a self-supervised prediction task using one of two architectures: Continuous Bag-of-Words (CBOW), which predicts a target word from its surrounding context, or Skip-gram, which performs the inverse prediction. The resulting word embeddings encode both syntactic and semantic structure in a dense vector space, enabling linear algebraic operations over meaning — most notably the well-known relationship $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$.

The significance of Word2Vec extends well beyond the paper itself. By demonstrating that semantic similarity can be captured as geometric proximity in a continuous vector space, it established the distributional hypothesis as an actionable engineering principle. This insight directly underlies the design of modern embedding models and vector databases, which are central components in Retrieval-Augmented Generation (RAG) pipelines [8]. In the context of this work, the quality and cost of the embedding stage in the Maestro framework's RAG pipeline is a direct consequence of choices made in this representational paradigm.

2.1.2 The Transformer Architecture

Despite the advances in distributed representations, sequence modelling remained constrained by recurrent architectures such as LSTMs and GRUs, which process tokens sequentially and suffer from vanishing gradients over long contexts. Vaswani et al. [18] proposed a radical departure: the Transformer, an architecture that dispenses with recurrence entirely and relies solely on attention mechanisms to model dependencies between all positions in a sequence simultaneously. The core operation, scaled dot-product attention, computes output representations as a weighted sum of value vectors, where weights are derived from the compatibility between query and key projections:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

By running multiple such attention operations in parallel — Multi-Head Attention — the model learns to simultaneously attend to different types of relationships across the input, such as syntactic dependencies and semantic co-references. Positional encodings are injected additively into the input embeddings to compensate for the architecture's inherent permutation invariance. The resulting model achieved 28.4 BLEU on the WMT 2014 English-to-German translation benchmark, surpassing all prior approaches while requiring a fraction of the training compute [18].

The Transformer's parallelism over sequence positions unlocked a new scaling regime for neural language models. Architectures derived from it — most notably encoder-only models such as BERT [3] for representation learning, and decoder-only models such as the GPT series [2] for text generation — became the standard building blocks of enterprise GenAI systems. Critically, the attention mechanism scales quadratically with sequence length, $O(n^2)$, meaning that context length is a primary cost driver in token-priced LLM APIs. This architectural property is foundational to the FinOps strategies explored in this dissertation, namely prompt compression, semantic caching, and dynamic model selection, as each seeks to reduce the effective n processed by the model or the frequency with which inference is invoked at all [11, 18].

Together, these two works define the representational substrate of modern GenAI: Word2Vec established that meaning can be encoded as position in a learned vector space, and the Transformer established that context-sensitive reasoning over such representations can be performed efficiently and at scale. The remainder of this chapter examines the architectural patterns, operational frameworks, and cost optimisation techniques that build upon these foundations in production-grade enterprise systems.

2.2 Large Language Models in Production

The preceding section established the two foundational breakthroughs that made modern language AI possible: distributed word representations and the Transformer architecture. Building upon these primitives, the field witnessed a rapid consolidation around a

unified paradigm — the large language model (LLM) — in which ever-larger Transformer-based networks, pre-trained on internet-scale corpora, acquire general-purpose language competence that can be redirected toward downstream tasks with minimal additional supervision. This section characterizes LLMs as a class, traces the methodological evolution from raw pre-training to instruction-following alignment, and examines the architectural and economic implications of deploying such systems at enterprise scale.

2.2.1 Large Language Models: An Overview

A large language model is a neural language model of sufficient parameter count and training data scale that general linguistic and reasoning capabilities emerge as a consequence of the pre-training objective alone, without task-specific architectural modification [2]. Pre-training proceeds via self-supervised next-token prediction over corpora of hundreds of billions to trillions of tokens, optimizing the model to assign high probability to naturally occurring text. The resulting parameter weights encode an implicit representation of syntax, semantics, world knowledge, and rudimentary reasoning chains that can be elicited through natural language prompting.

LLMs are predominantly realized as *decoder-only* Transformer architectures [18], in which the self-attention layers apply a causal mask that restricts each token's attention to its leftward context, preserving the autoregressive generation property. This design choice contrasts with the encoder-only variant typified by BERT [3], which produces bidirectional contextual representations optimized for discriminative tasks. For generative applications — intelligent assistants, automated reasoning, and document synthesis — the decoder-only paradigm dominates, and it is this class of model that underpins the enterprise GenAI systems examined in this dissertation.

A defining empirical finding in the development of LLMs is the existence of *scaling laws*: systematic, power-law relationships between model size, training compute, dataset volume, and downstream performance [2]. Scaling laws imply that performance is largely predictable and improvable by increasing any of the three axes, establishing a framework within which investment in compute translates into measurable capability gains. This property has driven the successive scaling of commercial models and, critically, has also made token consumption a dominant cost variable: larger models process the same input more expensively, directly motivating the cost optimization strategies central to this work.

2.2.2 From Pre-training to Instruction Tuning

Despite the generative competence acquired during pre-training, early LLMs exhibited a systematic misalignment between model behavior and user intent. A model trained solely to predict the next token on a web-scale corpus learns to continue text in statistically plausible ways, but not necessarily in ways that are *helpful*, *honest*, or *harmless* when deployed as an interactive assistant. Ouyang et al. [12] addressed this gap by introducing a three-stage fine-tuning pipeline centered on Reinforcement Learning from Human Feedback (RLHF).

The procedure operates as follows. First, a *supervised fine-tuning* (SFT) phase trains the base model on high-quality demonstrations of desired behavior, produced by human labelers on a curated prompt distribution. Second, a *reward model* is trained to predict human preference scores over pairs of model outputs, internalizing a proxy for human judgment. Third, the SFT model is further optimized against the reward model signal using Proximal Policy Optimization (PPO), an online reinforcement learning algorithm. The resulting InstructGPT models, despite comprising as few as 1.3 billion parameters, were consistently preferred by human evaluators over the unaligned 175-billion parameter GPT-3 baseline, demonstrating that alignment quality is not strictly a function of scale [12]. This result has profound practical implications: it established RLHF as the standard post-training procedure for production LLMs and confirmed that an aligned smaller model can match or exceed the utility of a larger unaligned one — a relationship that informs the dynamic model selection strategies explored in this dissertation.

2.2.3 Open-Weight Models and the Deployment Spectrum

The commercial LLM landscape is stratified between proprietary, API-accessed models and openly released, self-hostable alternatives. The former class, exemplified by OpenAI's GPT-4 family and Anthropic's Claude series, exposes inference through token-priced cloud APIs; the latter is represented most prominently by Meta AI's Llama series. Touvron et al. [17] introduced Llama 2, a family of models ranging from 7 to 70 billion parameters, pre-trained on two trillion tokens and released under a license permitting commercial use. Fine-tuned variants — Llama 2-Chat — further apply RLHF alignment, achieving performance competitive with closed-source alternatives on standard benchmarks while remaining locally deployable [17].

The existence of capable open-weight models introduces a consequential degree of freedom into enterprise GenAI architectures. Rather than committing uniformly to a single proprietary provider, organizations may route inference dynamically across a portfolio of models differentiated by capability tier, latency characteristics, and per-token cost. Smaller open-weight models, deployed on-premise or on reserved cloud instances, can serve high-volume, lower-complexity queries at substantially reduced marginal cost, while API-accessed frontier models handle tasks requiring peak reasoning performance. This model routing paradigm — often referred to as *LLM cascading* or *dynamic model selection* — is a central cost optimization mechanism evaluated in this dissertation in the context of the Maestro framework's multi-cloud deployment [17, 12].

Together, the progression from raw pre-training through RLHF alignment to the open-weight deployment spectrum defines the operational environment within which enterprise GenAI frameworks must function. The ability to select, combine, and switch between models of varying capability and cost is not merely an architectural convenience but a financial imperative in production systems where token consumption can scale three to five times per client without deliberate optimization [12].

2.3 Retrieval-Augmented Generation

Large language models store factual knowledge implicitly within their parameters, an approach that introduces well-documented limitations: parametric knowledge is frozen at training time, cannot be transparently audited, and tends to produce hallucinations when queries exceed the model's training distribution [8]. Retrieval-Augmented Generation (RAG) addresses these limitations by coupling a generative model with a dynamic, non-parametric memory that can be queried, updated, and inspected at inference time. The following subsections trace the theoretical foundations of this paradigm, survey its subsequent architectural evolution, and ground the discussion in the dense retrieval machinery that makes scalable semantic search tractable in practice.

2.3.1 Foundational Architecture: Lewis et al. (2020)

The canonical RAG formulation was introduced by Lewis et al. [8], who proposed a general-purpose fine-tuning recipe for generative models endowed with access to an external non-parametric memory. The architecture decomposes into two principal components: (i) a *retriever* $p_\eta(z \mid x)$, implemented as a Dense Passage Retriever (DPR) [6] whose bi-encoder architecture maps both queries and documents into a shared dense vector space via independent BERT encoders, and (ii) a *generator* $p_\theta(y_i \mid x, z, y_{1:i-1})$, implemented as BART-large [7], which conditions its output sequence on the concatenation of the original query x and retrieved passages z.

Formally, the top-K passages are selected by solving a Maximum Inner Product Search (MIPS) problem over a pre-indexed Wikipedia corpus of 21 million 100-word chunks:

$$p_\eta(z \mid x) \;\propto\; \exp\big(d(z)^\top q(x)\big)\,, \quad d(z) = \mathrm{BERT}_d(z),\; q(x) = \mathrm{BERT}_q(x). \tag{2.1}$$

Two marginalization strategies over the latent document variable z are proposed. In *RAG-Sequence*, a single retrieved document is assumed responsible for the entire output sequence:

$$p_{\text{RAG-Seq}}(y \mid x) \;\approx\; \sum_{z \in \text{top-}k} p_\eta(z \mid x) \prod_{i}^{N} p_\theta(y_i \mid x, z, y_{1:i-1}). \tag{2.2}$$

In *RAG-Token*, a distinct document may be selected at each generation step, enabling the generator to synthesize content from multiple retrieved passages:

$$p_{\text{RAG-Token}}(y \mid x) \;\approx\; \prod_{i}^{N} \sum_{z \in \text{top-}k} p_\eta(z \mid x)\, p_\theta(y_i \mid x, z, y_{1:i-1}). \tag{2.3}$$

The retriever and generator are trained end-to-end via minimization of the negative marginal log-likelihood $\sum_j -\log p(y_j \mid x_j)$, without any direct supervision on which documents should be retrieved. Crucially, the document encoder and its FAISS index are kept frozen during fine-tuning; only the query encoder and the BART generator are updated, substantially reducing computational cost.

Empirically, RAG established state-of-the-art results on three open-domain question answering benchmarks—Natural Questions (44.5 EM), WebQuestions (45.2 EM), and CuratedTrec (52.2 EM)—outperforming both purely parametric seq2seq baselines (e.g., T5-11B) and task-specific retrieve-and-extract systems (e.g., DPR-based readers) [8]. For generative tasks, human evaluators rated RAG outputs as more factual than a BART baseline in 42.7% of Jeopardy question generation trials, compared to only 7.1% in which BART was judged superior. An index hot-swapping experiment further demonstrated that updating the non-parametric memory by replacing the document index is sufficient to update the model's world knowledge without any retraining, a property of practical significance for production deployments.

2.3.2 Architectural Evolution: Naive, Advanced, and Modular RAG

Gao et al. [4] systematize the rapid evolution of RAG methods into a three-tier taxonomy, providing a structural lens through which the design of contemporary systems can be evaluated.

Naive RAG. The earliest post-ChatGPT RAG implementations follow a minimal *Retrieve-then-Read* pipeline consisting of three sequential steps: (1) *indexing*, in which raw documents are chunked, encoded via an embedding model, and stored in a vector database; (2) *retrieval*, in which the top-K chunks most similar to the user query are identified via cosine similarity over dense embeddings; and (3) *generation*, in which the query and retrieved chunks are concatenated into a prompt supplied to the language model [4]. While effective in simple scenarios, Naive RAG exhibits well-characterized failure modes: low retrieval precision due to semantic gaps between queries and documents, context window saturation from uninformative passages, and hallucinations that arise when the generator defaults to parametric knowledge in the presence of noisy retrieved context.

Advanced RAG. Advanced RAG introduces targeted interventions at both the pre-retrieval and post-retrieval stages to address the precision and coherence limitations of the naive approach [4]. Pre-retrieval strategies include query rewriting, query expansion into sub-questions, and the construction of *Hypothetical Document Embeddings* (HyDE), in which the model first generates a hypothetical answer and then uses its embedding to retrieve semantically closer passages. Post-retrieval strategies include cross-encoder reranking to elevate the most relevant passages and context compression to eliminate redundant or off-topic content prior to generation. Indexing enhancements—such as sliding-window chunking, fine-grained segmentation, and metadata attachment—further improve retrieval recall and temporal awareness. These modifications maintain the linear Retrieve-then-Read structure of Naive RAG while substantially improving end-to-end generation quality.

Modular RAG. Modular RAG supersedes the fixed-pipeline paradigm by decomposing the system into independently composable functional modules: *Search*, *Memory*, *Routing*,

Predict, and *Task Adapter* [4]. This architecture supports non-sequential execution patterns, including iterative retrieval loops (e.g., ITER-RETGEN [14]), adaptive retrieval triggered by generation uncertainty (e.g., FLARE [5]), and self-reflective retrieval with token-level reflection signals (e.g., Self-RAG [1]). The modular design also enables seamless integration with complementary techniques—fine-tuning the retriever, fine-tuning the generator, or training both collaboratively—without requiring architectural changes to the pipeline. Gao et al. [4] position Modular RAG as the current frontier, noting that its flexibility allows task-specific orchestration patterns that static pipelines cannot accommodate.

A practical implication of this taxonomy is that RAG and fine-tuning are not mutually exclusive. RAG excels in dynamic environments where knowledge must be updated frequently and interpretable provenance is required, whereas fine-tuning is better suited for scenarios demanding stylistic adaptation or the internalization of domain-specific reasoning patterns. In knowledge-intensive tasks, Gao et al. [4] report that RAG consistently outperforms unsupervised fine-tuning, both on knowledge encountered during pre-training and on entirely novel factual information.

2.3.3 Dense Semantic Retrieval: Sentence-BERT

The retrieval component of any RAG pipeline depends critically on the quality and computational tractability of sentence-level embeddings. Vanilla BERT representations are unsuitable for large-scale semantic search: because BERT processes sentence pairs jointly through a cross-attention encoder, identifying the most similar pair in a corpus of n sentences requires $O(n^2)$ inference passes, which Reimers and Gurevych [13] report as approximately 65 hours for $n = 10{,}000$ sentences. Furthermore, averaging BERT token embeddings or using the `[CLS]` token output yields representations that perform worse on semantic textual similarity (STS) tasks than simple GloVe embeddings, owing to BERT's lack of explicit metric learning for sentence-level geometry.

Reimers and Gurevych [13] address these limitations by introducing *Sentence-BERT* (SBERT), a modification of BERT using Siamese and triplet network structures to produce fixed-size, semantically meaningful sentence embeddings amenable to efficient cosine-similarity comparison. The architecture passes each sentence independently through a shared BERT encoder, then applies mean-pooling over the output token embeddings to obtain a single vector representation $\mathbf{u}$ and $\mathbf{v}$ for a sentence pair. Three training objectives are supported depending on the available supervision signal:

- **Classification objective** (for NLI corpora): a softmax classifier over the concatenation $[\mathbf{u};\ \mathbf{v};\ |\mathbf{u} - \mathbf{v}|]$ trained to predict entailment, contradiction, or neutral labels.

- **Regression objective** (for STS corpora): cosine-similarity between $\mathbf{u}$ and $\mathbf{v}$ optimized with mean-squared error against human-annotated similarity scores.

- **Triplet objective**: a margin-based ranking loss that ensures the anchor embedding $\mathbf{a}$

is closer to a positive $\mathbf{p}$ than to a negative $\mathbf{n}$ by a margin ϵ:

$$\mathcal{L}_{\text{triplet}} = \max(\|\mathbf{a} - \mathbf{p}\| - \|\mathbf{a} - \mathbf{n}\| + \epsilon,\ 0)\,. \tag{2.4}$$

The Siamese design decouples encoding from comparison: document embeddings are computed once at indexing time and stored in a vector index, reducing inference at query time to a single forward pass followed by a nearest-neighbor lookup. This reduces the 65-hour pairwise comparison to approximately 5 seconds for a 10,000-sentence corpus, while preserving—or improving upon—BERT's accuracy on STS benchmarks [13]. On the STS tasks evaluated, SBERT achieves an improvement of 11 Spearman correlation points over InferSent and 5 points over Universal Sentence Encoder.

2.3.4 Connection to LlamaIndex

The theoretical contributions reviewed above map directly onto the abstractions provided by LlamaIndex, the framework employed in this work. LlamaIndex operationalizes the RAG pipeline of Lewis et al. [8] through a modular architecture that cleanly separates ingestion, indexing, retrieval, and synthesis [9]. During the *ingestion* stage, `SimpleDirectoryReader` loads raw documents, which are subsequently segmented by `SentenceSplitter` into overlapping chunks—a direct implementation of the chunking strategies discussed by Gao et al. [4] for Advanced RAG. The resulting nodes are encoded via a configurable embedding model (`Settings.embed_model`), which defaults to OpenAI's `text-embedding-3-small` but is readily substituted with any SBERT-compatible model from the HuggingFace ecosystem (e.g., `BAAI/bge-small-en-v1.5`), whose architecture descends directly from the Siamese network paradigm of Reimers and Gurevych [13].

At query time, LlamaIndex's `VectorStoreIndex` retrieves the top-K most relevant nodes via cosine similarity—the same MIPS formulation as DPR in [8]—and passes the retrieved context together with the user query to the configured language model for response synthesis. The framework further exposes advanced retrieval patterns consistent with the Modular RAG taxonomy of Gao et al. [4]: `RouterQueryEngine` implements semantic routing to multiple sub-indexes; post-retrieval reranking modules (e.g., `SentenceTransformerRerank`) apply cross-encoder scoring to improve precision; and iterative retrieval agents can be composed from LlamaIndex's `ReActAgent` infrastructure. In this sense, LlamaIndex provides a production-grade instantiation of the entire RAG design space surveyed by Gao et al. [4], grounded in the dense retrieval foundations established by Lewis et al. [8] and Reimers and Gurevych [13].

2.4 Related Work

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A

Appendix Title

B

Appendix Title